

Model Compression & Demographic Fairness in Face Verification

Dissertation Progress — Experimental Findings & Methodology

Abbreviations

- **MTCNN**: Multi-task Cascaded Convolutional Networks (face detector)
- **SCRFD**: Selective Convolutional Region Face Detector (face detector)
- **ONNX**: Open Neural Network Exchange (model format)
- **ORT**: ONNX Runtime (inference engine)
- **INT8**: 8-bit integer quantisation (compressed weights)
- **QDQ**: Quantize-Dequantize (quantisation pattern: FP32 → INT8 → compute → FP32)
- **FP32**: 32-bit floating point (full precision, uncompressed)
- **CPU/GPU**: Processor types (latency measurements on CPU)
- **FMR**: False Match Rate (incorrectly accepting two different people as the same)
- **FNMR**: False Non-Match Rate (incorrectly rejecting the same person twice)
- **FDR**: Fair Difference Ratio (gap between demographic subgroups' error rates)
- **LFW**: Labeled Faces in the Wild (benchmark dataset, 13,233 images)
- **RQ**: Research Question (RQ1, RQ2, RQ3)
- **QA**: Quality Assurance (accuracy-only checking, without fairness audits)
- **buffalo_l** / **buffalo_s**: InsightFace model packs (large and small variants)
- **w600k_r50**: ResNet-50 trained on WebFace600K with ArcFace loss
- **w600k_mbf**: MobileFaceNet trained on WebFace600K with ArcFace loss

Methodology

Pipeline: MTCNN/SCRFD detect → ArcFace ONNX embed (InsightFace) → cosine similarity → LFW 10-fold eval with per-subgroup FMR/FNMR (LFW Attributes) and FDR (Fair Difference Ratio). Only the model artefact changes between runs; evaluation logic is never modified. All latency measurements are median of 3 clean runs on CPU (CPUExecutionProvider), clean system state, n_warmup=100.

Two backbones evaluated: ResNet-50 (buffalo_l, w600k_r50.onnx) and MobileFaceNet (buffalo_s). Three compression states per backbone: FP32 baseline, structured pruning (ResNet-50: layer4-only 40%; MobileFaceNet: last-4-blocks 15%), and INT8 static QDQ quantisation (MobileFaceNet with depthwise layers excluded from quantisation to avoid ORT CPU kernel gap).

RQ1 — Does Compression Widen Demographic Fairness Gaps? (ResNet-50)

Metric	FP32	Pruned (40%, layer4)	INT8 (static QDQ)
Accuracy	99.83%	99.53%	99.82%

FDR_FMR	0.63%	0.63%	0.63%
FDR_FNMR	0.95%	1.26%	0.95%
Latency (median, ms)	26.69	25.93	14.09

Finding: Pruning widens the FNMR fairness gap by 33% (0.95% → 1.26%); INT8 quantisation shows no widening. FMR gap remains flat across all variants (0.63%). Compression methods are not fairness-equivalent, even when accuracy is similar.

Caveat: Black-subgroup FNMR change (0% → 0.65%) reflects a single additional error on n=155 genuine pairs—flagged as a small-sample limitation.

RQ2 — Is MobileFaceNet Inherently Fairer Under Compression?

Metric	FP32	Pruned (15%, last-4)	INT8 (depthwise-excluded)
Accuracy	99.57%	99.30%	99.60%
FDR_FMR	0.63%	0.67%	0.63%
FDR_FNMR	0.95%	5.32%	0.63%
Latency (median, ms)	3.33	3.78	5.31

Finding: No. MobileFaceNet is less fairness-robust under compression than ResNet-50. Pruning causes a 4x increase in FDR_FNMR (0.95% → 5.32%), compared to ResNet-50's 1.3x (0.95% → 1.26%). The accuracy drop is small (0.27pp), but the fairness harm is substantial. An accuracy-only QA check would have missed this.

Note on RQ2 baseline: MobileFaceNet FP32 FDR values are identical to ResNet-50 FP32. This was verified as genuine (not a pipeline bug): both models fail the same 5 hard LFW pairs in the Asian subgroup—a known LFW phenomenon with extreme pose/lighting changes.

RQ3 — Does Quality Filtering Explain the Gaps Away? (FaceQNet)

FaceQNet v1 (official weights, ResNet-50 backbone) run on SCRFD-aligned 224×224 crops; validated against authors' precomputed LFW scores (7/8 spot-checks matched within 0.01–0.07). Percentile-based filtering: bottom 10%/20%/30% of quality scores excluded.

Confound found: Filtering removes the 'Unknown' demographic subgroup (images lacking race annotation) at 2–4× the rate of White/Black/Asian (34.3%/51.4%/68.6% removed at bottom-10%/20%/30% vs ~10% for others). Any apparent fairness 'improvement' for that subgroup is a data-loss artefact, not evidence.

Result: mostly ruled out. Only one case showed a large shrink under filtering (MobileFaceNet Pruned: 5.32% → 0.29%), traced to a single failed pair in a 17-image subgroup—not a general effect. Fairness gaps seen under compression are a real property of the compression methods, not a data-quality artefact filtering can explain away.

Key Technical Findings

- **Full-network pruning collapses accuracy:** Full-network structured pruning on ResNet-50 collapsed accuracy to 61% because conv2 in each IResNet block feeds the residual Add with no BatchNorm buffer—error compounds multiplicatively across 24 blocks. Fix: restrict pruning to layer4 (closest to output); tolerates 40% sparsity with only 0.3pp accuracy loss.
- **INT8 opset upgrade required:** Static QDQ quantisation needed ONNX opset 11→13 upgrade for per-channel Q/DQ op support. Model size reduction: 174MB → 44MB (ResNet-50), negligible accuracy loss.
- **Depthwise conv quantisation kernel gap:** MobileFaceNet INT8 was initially 10× slower than FP32 (33.44ms vs 3.46ms). ORT profiling confirmed: no efficient CPU kernel for depthwise/grouped conv QLinearConv (606µs/call vs 40µs/call for pointwise). Fix: exclude depthwise layers from quantisation; recovered 5.26ms latency at no accuracy/fairness cost. This is a citable architecture/quantisation-backend interaction.
- **Latency benchmarking sensitivity:** Absolute latency figures are sensitive to system load. Original ResNet-50 INT8 p95 (62.1ms) was diagnosed as background process contention, not an ORT issue. All reported figures are from clean-session re-runs (background apps closed, High Performance power plan).

Limitations & Caveats

- LFW subgroup sizes are small (Black n=155 genuine, Asian n=318, Unknown n=17). Single-pair differences swing FDR by ~0.3–0.6%, so subgroup-level trends should be read qualitatively except where large and traceable.
- MobileFaceNet pruning sensitivity sweep was in-sample optimistic vs held-out result (predicted 98.8–99.1%, actual 99.30%), so final config used a safety margin rather than exact prediction.
- FaceQNet quality filtering analysis is complicated by asymmetric subgroup removal rates—'Unknown' subgroup results should not be treated as reliable.
- Latency measurements are CPU-only (ONNX Runtime CPUExecutionProvider). GPU deployment patterns may differ.

Next Steps

All three RQs now have data and are defensible. Ready to write the full dissertation: literature review, methodology chapter, results, and discussion. The main contribution is the compression-fairness audit methodology and the finding that accuracy-only QA misses fairness harm under compression.